

Scalability & Future Plan

Date: 30 September 2026
Team ID: SWTID-2026-3656
Project Name: EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks: 1 Mark

Current Limitations

No.	Limitation	Impact	Priority
1	Local model is large	First explanation request needs disk/download	High
2	Gemini provider dependency	Cloud features depend on key/network/quota	High
3	No persistent progress data	No history or learner analytics	Medium

Scalability Plan

Aspect	Current State	Proposed Upgrade
User load	Single FastAPI process	Production workers and autoscaling
Data storage	No database	PostgreSQL for profiles, history, progress
Performance	Cached local model, synchronous requests	Async jobs, queues, response caching
Security	<code>.env</code> key	Secret manager, authentication, rate limits, audit logs

Future Roadmap

Phase	Enhancement	Impact
2	Voice and multilingual learning	Improved accessibility
3	Image/PDF question solving	Richer study inputs
4	Progress dashboard and gamification	Motivation and retention
5	Teacher/parent dashboards and LMS integration	Institutional adoption